Technical Implementation Plan — Employee Attendance System

1. Overview

An employee attendance management system with two frontend apps (Admin & Employee) backed by a NestJS API, MySQL database, Kafka message broker for event-driven audit logging and real-time notifications via SSE.

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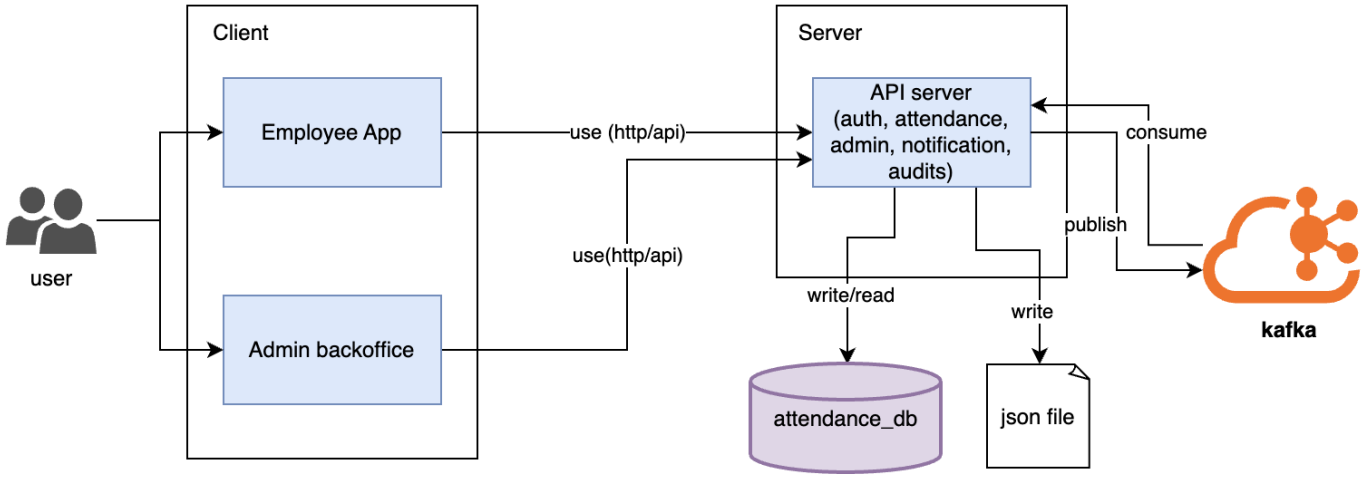
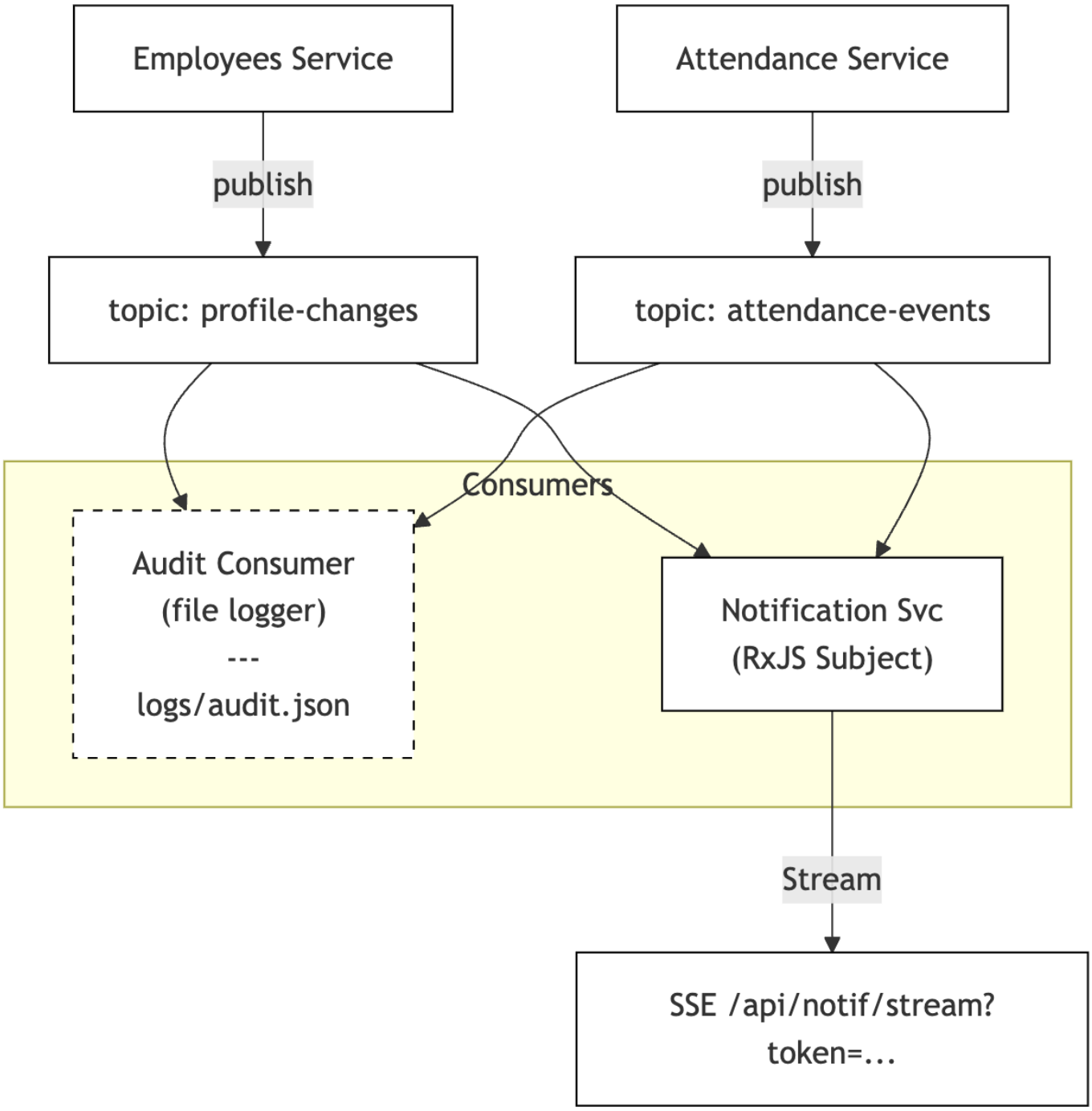
2. System Architecture



Event Flow (Kafka)



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3. Tech Stack

Layer	Technology	Version
Backend	NestJS (Node.js)	11.x
ORM	TypeORM	latest
Database	MySQL	8
Message Broker	Apache Kafka (KRaft, no Zookeeper)	3.7.0
Auth	Passport + JWT (@nestjs/jwt)	11.x
Frontend	React + TypeScript	19.x

Layer	Technology	Version
Bundler	Vite	8.x
CSS	Tailwind CSS	4.x
HTTP Client	Axios	1.x
Infra	Docker Compose	v2

4. API List

4.1 Auth

Method	Endpoint	Auth	Description
POST	`/api/auth/login`	Public	Login with email/password. Returns `accessToken`, `refreshToken`, employee data.
POST	`/api/auth/refresh`	Public	Exchange refresh token for new access token.

4.2 Employees

Method	Endpoint	Auth	Description
GET	`/api/employees/me`	JWT (any)	Get current employee's profile.
PATCH	`/api/employees/me`	JWT (any)	Update own profile (phone, password, photo). Publishes `profile-changes` Kafka event.
GET	`/api/employees`	JWT (admin)	List employees with pagination & search.
POST	`/api/employees`	JWT (admin)	Create new employee.
PATCH	`/api/employees/:id`	JWT (admin)	Admin update employee fields.

4.3 Attendance

Method	Endpoint	Auth	Description
POST	`/api/attendance/clock-in`	JWT (any)	Clock in. Multiple clock-ins per day allowed. Publishes `attendance-events` Kafka event.
POST	`/api/attendance/clock-out`	JWT (any)	Clock out the latest active session. Publishes `attendance-events` Kafka event.
GET	`/api/attendance/today`	JWT (any)	Get all attendance records for today (array).

Method	Endpoint	Auth	Description
GET	`/api/attendance/summary`	JWT (any)	Get attendance summary for date range. Returns unique day counts. Query: `?from=YYYY-MM-DD&to=YYYY-MM-DD`
GET	`/api/attendance/all`	JWT (admin)	List all attendance records with filters. Query: `?page=&limit=&from=&to=&employee_id=`

4.4 Notifications

Method	Endpoint	Auth	Description
GET	`/api/notifications/stream`	Token (query)	SSE stream for real-time notifications. Admin only. Query: `?token=JWT`

5. Data Model

EMPLOYEES			
int	id	PK	auto-increment
varchar	name		NOT NULL
varchar	email		NOT NULL, UNIQUE
varchar	password		NOT NULL, excluded from SELECT
varchar	photo_url		nullable
varchar	position		NOT NULL
varchar	phone		nullable
enum	role		employee | admin
boolean	is_active		default: true
timestamp	created_at		auto
timestamp	updated_at		auto

ATTENDANCES			
int	id	PK	auto-increment
int	employee_id	FK	NOT NULL, ON DELETE CASCADE
date	date		NOT NULL
timestamp	clock_in		nullable
timestamp	clock_out		nullable
enum	status		masuk | pulang
timestamp	created_at		auto
timestamp	updated_at		auto

Notes:

- One employee can have **multiple attendance records per day** (multiple clock in/out sessions).
- `employee_id + date` is a **non-unique index** (not a unique constraint).
- Deleting an employee **cascades** to all their attendance records.
- Audit log is file-based (`logs/audit.json`), not a database table.

5.1 Audit Log (file-based)

Stored at `backend/logs/audit.json`. Each entry:

```
{
  "action": "clock_in | clock_out | profile_updated",
  "entity": "attendance | employee",
  "entityId": 1,
  "employeeId": 1,
  "employeeName": "John Doe",
  "changes": {},
  "timestamp": "2026-04-02T07:00:00.000Z"
}
```

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6. Authentication & Authorization

Aspect	Implementation
Strategy	JWT via Passport (`passport-jwt`)
Access Token	15 minute expiry, sent in `Authorization: Bearer <token>`
Refresh Token	7 day expiry, exchanged via `/api/auth/refresh`
Password Hashing	bcrypt (salt rounds: 10)
Role Guard	Custom `RolesGuard` + `@Roles('admin')` decorator
SSE Auth	JWT passed as query parameter `?token=` (browser EventSource limitation)

7. Kafka Topics & Consumer Groups

Topic	Producer	Consumers
`profile-changes`	EmployeesService	AuditConsumer (`audit-consumer-group`), NotificationsService (`notification-consumer-group` — forwards `profile_updated`)
`attendance-events`	AttendanceService	AuditConsumer (`audit-consumer-group-attendance`), NotificationsService (`notification-consumer-group-attendance` — filters: only `late_clock_in` and `early_leave` forwarded as notifications; normal clock in/out is audit-only)

Each consumer uses a separate KafkaJS consumer instance because KafkaJS requires `run()` to be called once per consumer.

8. Design Patterns

Pattern	Where	How
Observer	NotificationsService	RxJS `Subject` acts as observable; SSE clients are observers subscribing via `getNotificationStream()`.
Event-Driven	Kafka pipeline	Services publish domain events; consumers react asynchronously. Decouples audit/notification from business logic.
Repository	TypeORM	`Repository<Entity>` injected via `@InjectRepository()`.
Guard/Decorator	Auth	`JwtAuthGuard` + `RolesGuard` + `@Roles()` for declarative access control.
DTO Validation	Controllers	`class-validator` + `ValidationPipe(whitelist, transform)` for input sanitization.

Pattern	Where	How
Global Filter	HttpExceptionFilter	Catches all exceptions, returns consistent error response format.

9. Frontend Architecture

9.1 Employee App (`:5173`)

Route	Page	Description
`/login`	LoginPage	Employee login
`/attendance`	LiveAttendancePage	Live clock with clock in/out buttons, today's log
`/summary`	AttendanceSummaryPage	Attendance history with date range
`/profile`	ProfilePage	View/edit own profile

9.2 Admin App (`:5174`)

Route	Page	Description
`/login`	LoginPage	Admin login
`/employees`	EmployeeListPage	Employee list with search/pagination
`/employees/new`	EmployeeFormPage	Create employee
`/employees/:id`	EmployeeFormPage	Edit employee
`/attendance`	AttendanceMonitorPage	Real-time attendance feed via SSE

10. Key Business Rules

1. **Multiple clock in/out per day**: Employees can clock in and out multiple times in a single day. Each clock-in creates a new attendance record.
2. **Clock out targets active session**: Clock out finds the latest record with `clock_out IS NULL` for today.
3. **Summary counts unique days**: `totalDays` and `totalPresent` use `Set<date>` to deduplicate — 3 sessions in 1 day = 1 day counted.
4. **Profile changes are event-sourced**: Any profile update (phone, password, photo) emits a Kafka event with the list of changed fields.
5. **Admin-only SSE**: Notification stream requires admin JWT.
6. **Photo upload**: Max 5MB, image types only (jpg, jpeg, png, gif, webp). Stored on disk at `backend/uploads/`.

10.1 HR Notification Rules

Admin notifications are **filtered for actionable events only** — routine clock in/out does not generate a push notification (still recorded in audit log).

Notification Type	Trigger	Severity	Delivery
`late_clock_in`	Employee clocks in after 09:15 (configurable)	High	Real-time SSE
`early_leave`	Employee clocks out before 15:00 (configurable)	High	Real-time SSE
`absent_alert`	Cron at 10:00 weekdays — employees with no clock-in	High	Scheduled SSE
`forgot_clock_out`	Cron at 19:00 weekdays — records with `clock_out IS NULL`	High	Scheduled SSE
`profile_updated`	Employee updates profile data	Medium	Real-time SSE
Normal clock in/out	—	—	Audit log only

Configuration constants (in `notifications.service.ts`):

Constant	Default	Description
`WORK_START_HOUR`	9	Expected clock-in hour
`WORK_START_MINUTE`	0	Expected clock-in minute
`LATE_THRESHOLD_MINUTES`	15	Minutes after work start = late
`EARLY_LEAVE_HOUR`	15	Clock-out before this hour = early leave

Cron schedule (via `@nestjs/schedule`):

- `0 10 * * 1-5` — Absent check (Mon–Fri at 10:00)
- `0 19 * * 1-5` — Forgot clock-out check (Mon–Fri at 19:00)

Frontend severity styling: High-severity notifications show a red left border; medium shows yellow. Absent/forgot alerts display up to 3 employee names with "+N lainnya" overflow.

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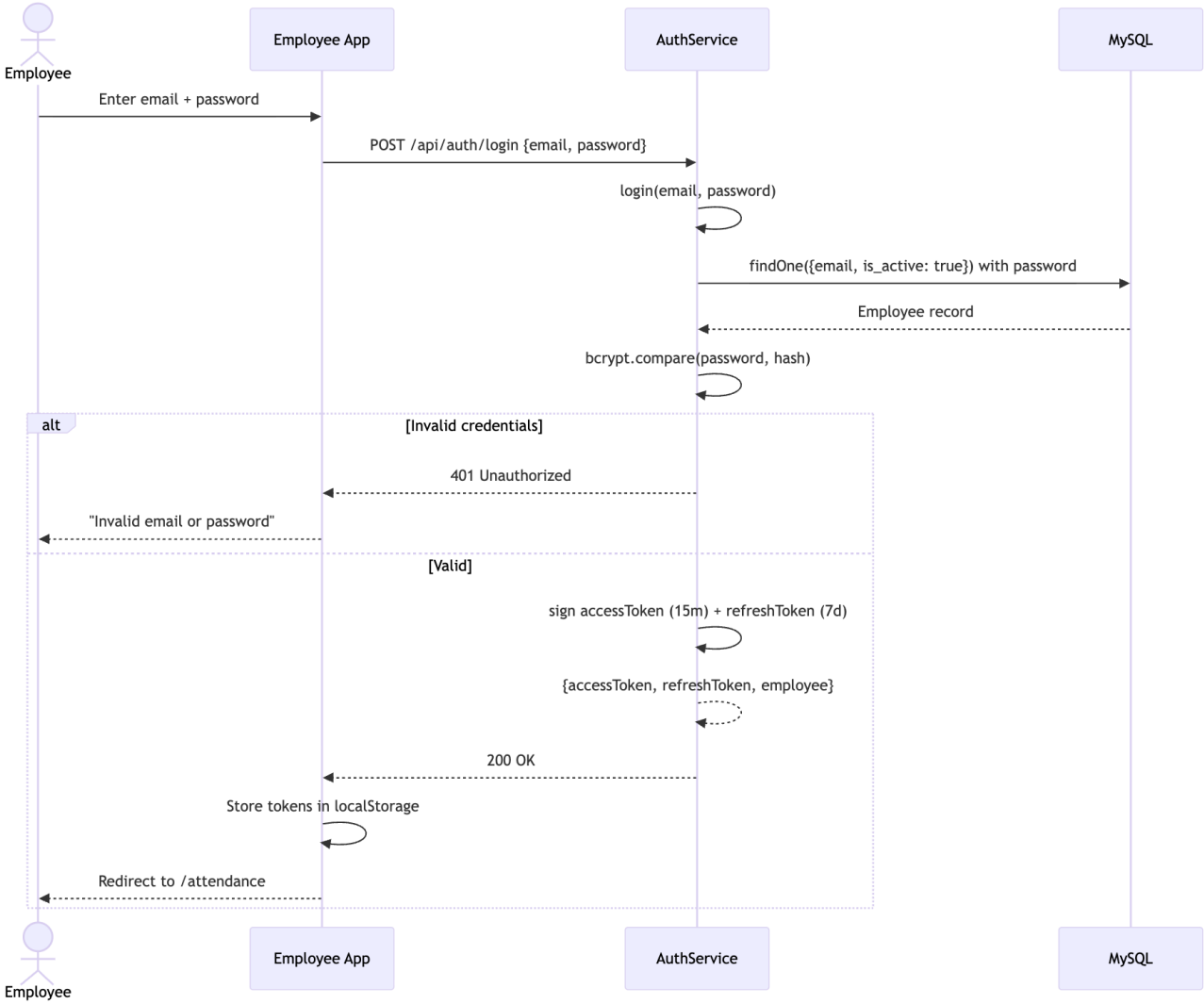
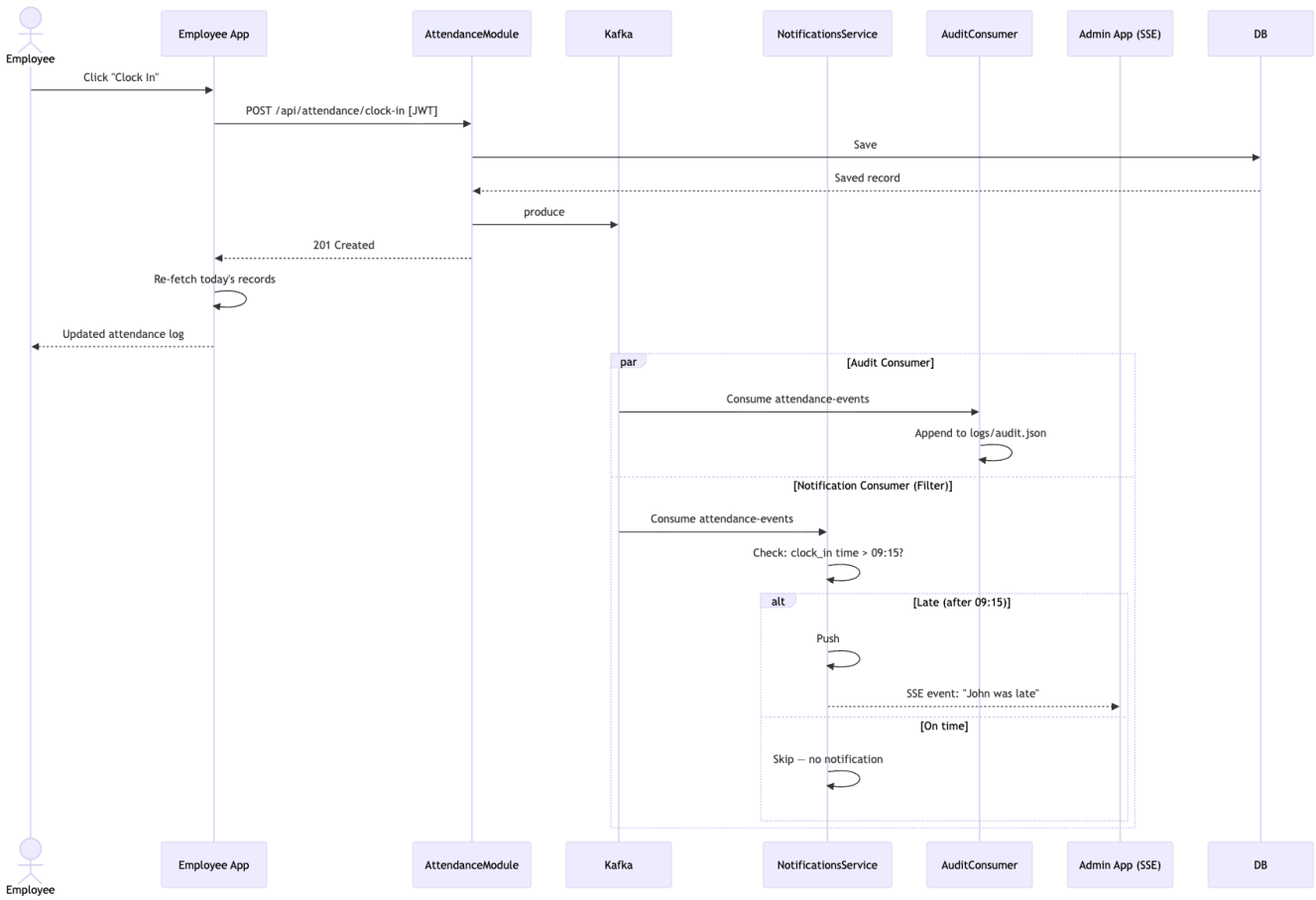
11. Sequence Diagrams

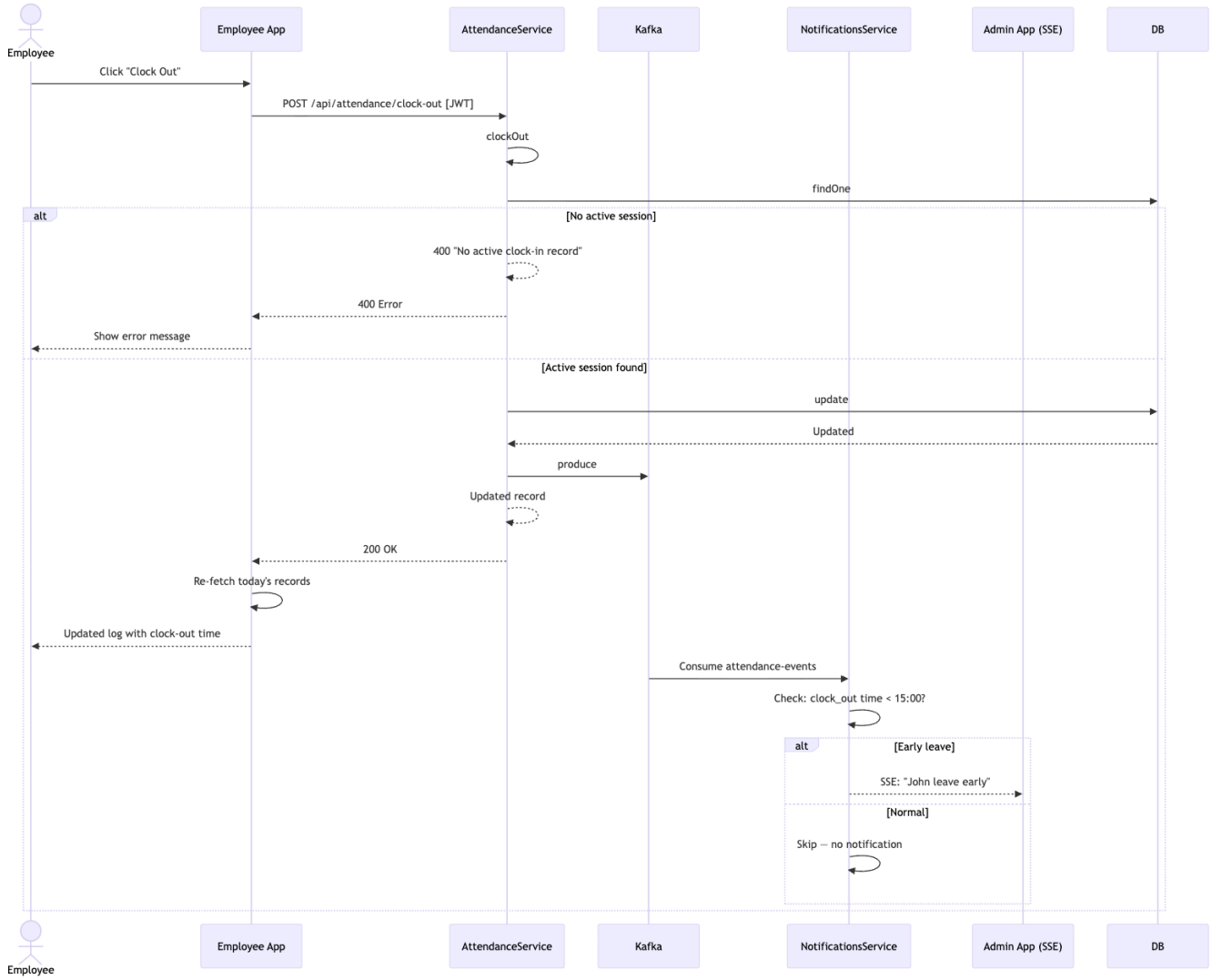
11.1 Employee Login



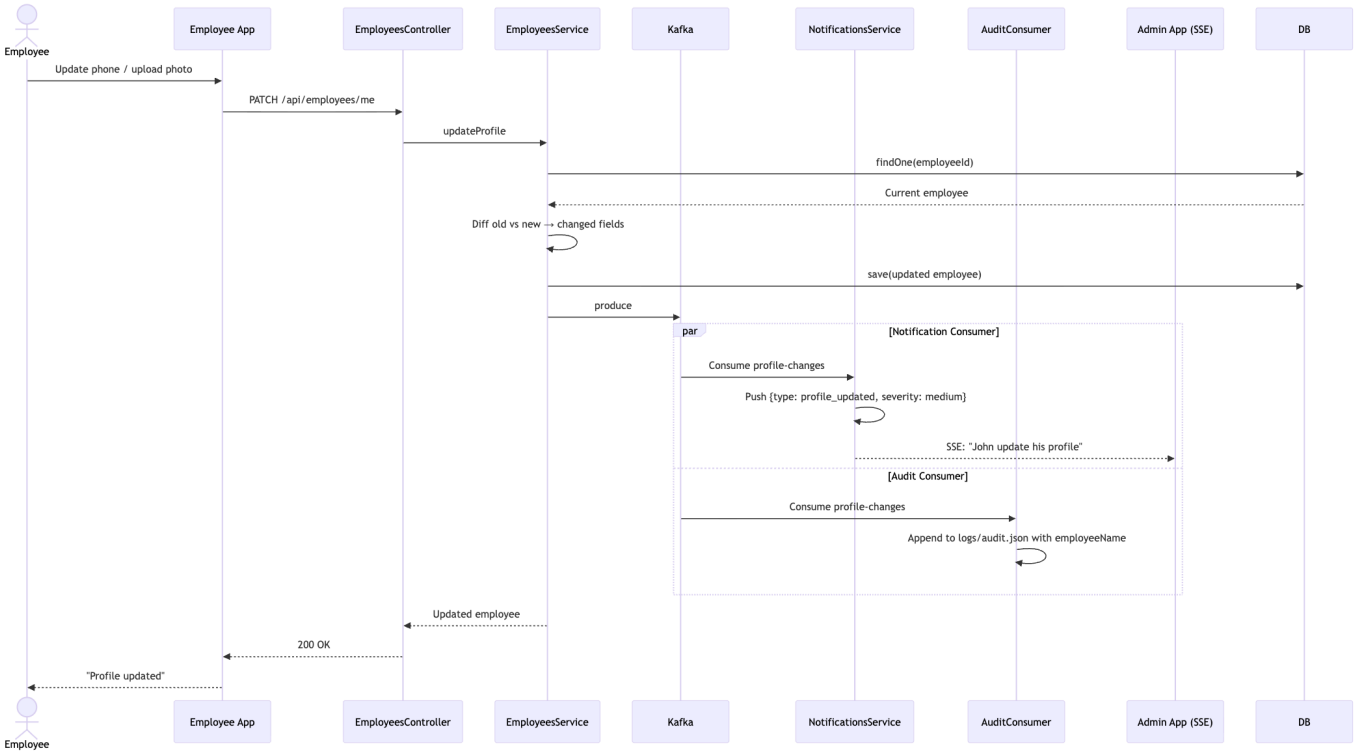
11.2 Clock In (Multiple per Day + Late Detection)



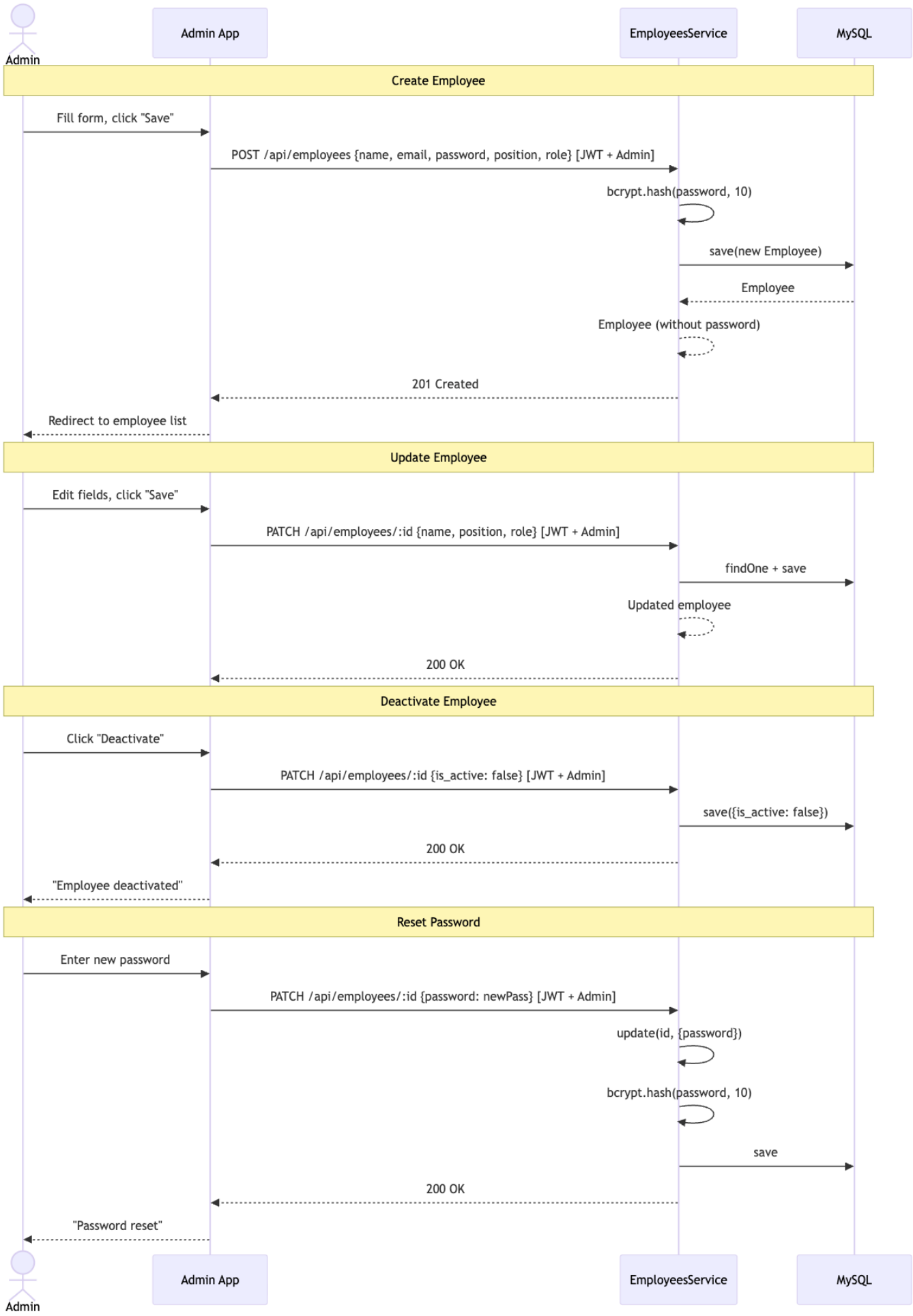
11.3 Clock Out (Active Session + Early Leave Detection)



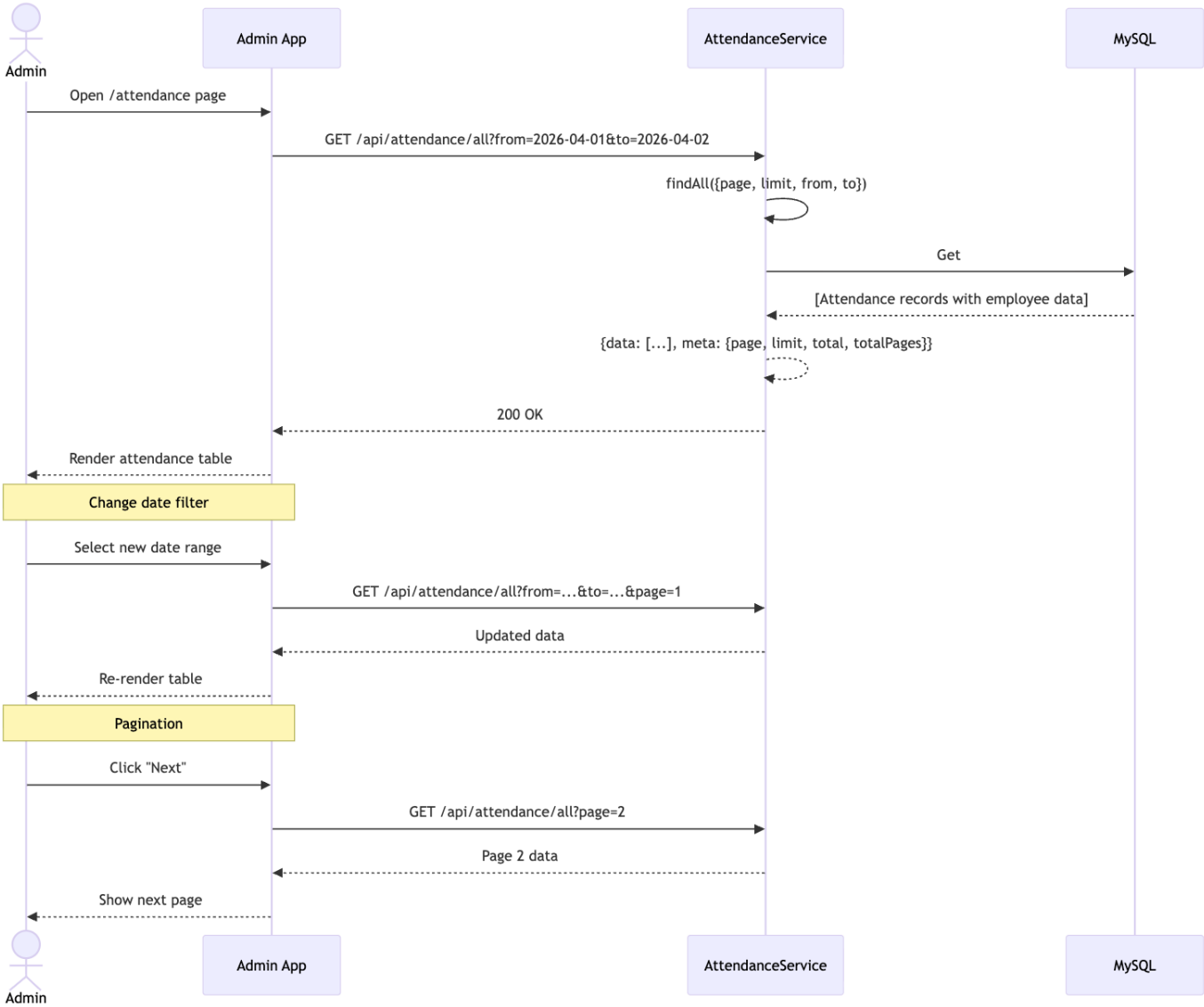
11.4 Profile Update (Kafka Fan-Out)



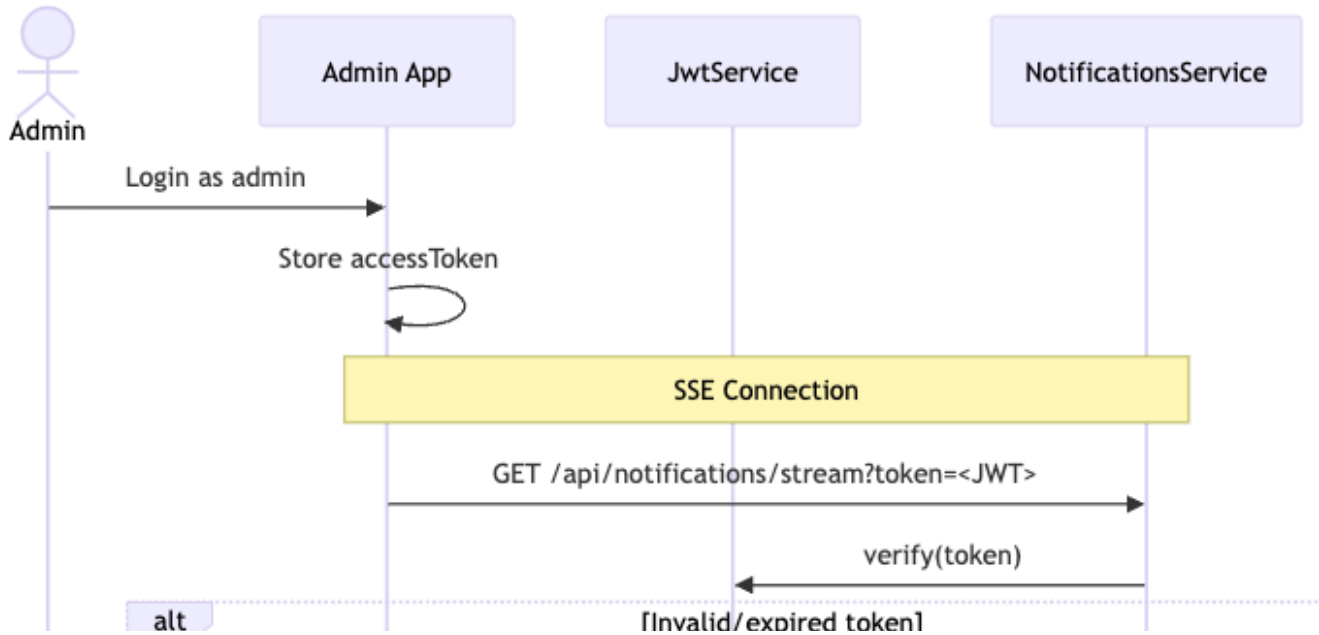
11.5 Admin — Employee Management (CRUD)

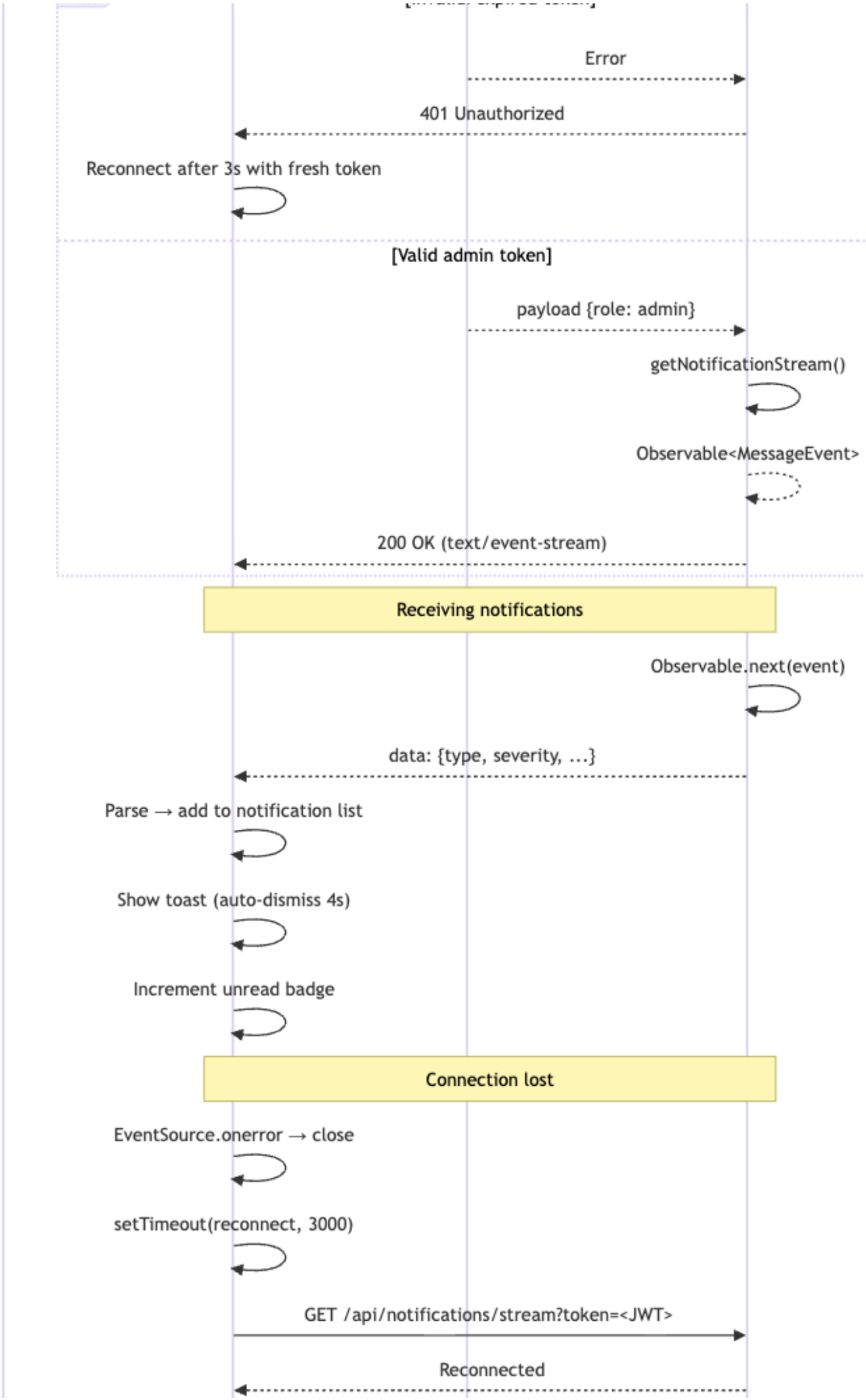


11.6 Admin — Attendance Monitoring



11.9 SSE Connection & Admin Notification Flow




Error
401 Unauthorized
Reconnect after 3s with fresh token
[Valid admin token]
payload {role: admin}
getNotificationStream()
Observable<MessageEvent>
200 OK (text/event-stream)
Receiving notifications
Observable.next(event)
data: {type, severity, ...}
Parse → add to notification list
Show toast (auto-dismiss 4s)
Increment unread badge
Connection lost
EventSource.onerror → close
setTimeout(reconnect, 3000)
GET /api/notifications/stream?token=<JWT>
Reconnected

12. Infrastructure (Docker Compose)

Service	Image	Port	Notes
mysql	mysql:8	3306	Persistent volume `mysql_data`. Init script creates `attendance_db` and `audit_db`.
kafka	apache/kafka:3.7.0	9092	KRaft mode (no Zookeeper). Auto-create topics enabled.

Backend and frontends run locally (not containerized).

13. Assumptions

1. **Single instance deployment**: No horizontal scaling considerations. Kafka consumer groups are per-module, not per-instance.
2. **File-based audit log**: Acceptable for current scale. Not suitable for production — would need a dedicated audit DB or log aggregation service.
3. **TypeORM synchronize: true**: Schema auto-sync in non-production. Production should use migrations.
4. **No rate limiting**: Clock in/out endpoints have no throttling. Assumes trusted internal network.
5. **CORS open**: `app.enableCors()` with no origin restriction. Should be locked down for production.
6. **JWT secret fallback**: `process.env.JWT_SECRET || 'fallback-secret'` — the fallback must never be used in production.
7. **No soft delete**: Employee deletion cascades to attendance records.
8. **No timezone handling**: Dates use server timezone (`new Date().toISOString().split('T')[0]`).
9. **SSE token via query param**: Required because browser `EventSource` API doesn't support custom headers.

14. Rationale & Trade-offs

Decision	Rationale
Kafka over simple webhooks	Decouples producers from consumers. Audit and notification can evolve independently. Kafka provides durability and replay.
RxJS Subject for SSE	Native NestJS `@Sse()` support. Subject allows multicasting to all connected clients with zero config.
Two separate frontend apps	Admin and employee have different layouts, routes, and access patterns. Separate bundles keep each app lean.

Decision	Rationale
File-based audit vs DB	Simplicity for dev/demo. Append-only JSON log is easy to inspect. Clear upgrade path to DB-backed audit.
Non-unique attendance index	Allows multiple clock in/out sessions per day. Business logic enforces ordering (clock out before next clock in).
Tailwind CSS	Utility-first approach speeds up UI development. No custom CSS maintenance.
Vite over Webpack	Faster HMR, simpler config, native ESM support. Standard choice for React + TypeScript in 2026.

15. Project Structure

```
dexa/
├── docker-compose.yml          # MySQL + Kafka
├── init-db.sql                 # DB initialization
├── backend/
│   └── src/
│       ├── main.ts             # Bootstrap, global pipes/filters
│       ├── app.module.ts       # Root module
│       ├── config/             # Database config
│       ├── common/             # Filters, interceptors, decorators
│       ├── auth/               # Login, JWT, guards
│       ├── employees/          # Employee CRUD, profile
│       ├── attendance/         # Clock in/out, summary
│       ├── kafka/              # Producer & consumer services
│       ├── audit/              # Kafka → file audit log
│       ├── notifications/      # Kafka → SSE (Observer pattern)
│       └── seeds/              # DB seeder
├── frontend/
│   ├── admin-app/              # React admin dashboard
│   └── employee-app/           # React employee self-service
└── docs/
    └── TECHNICAL_IMPLEMENTATION_PLAN.md
```